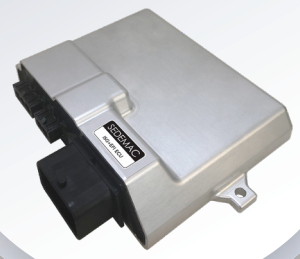
SEDEMAC™

Sensorless Integrated Starter Generator (ISG) ECU

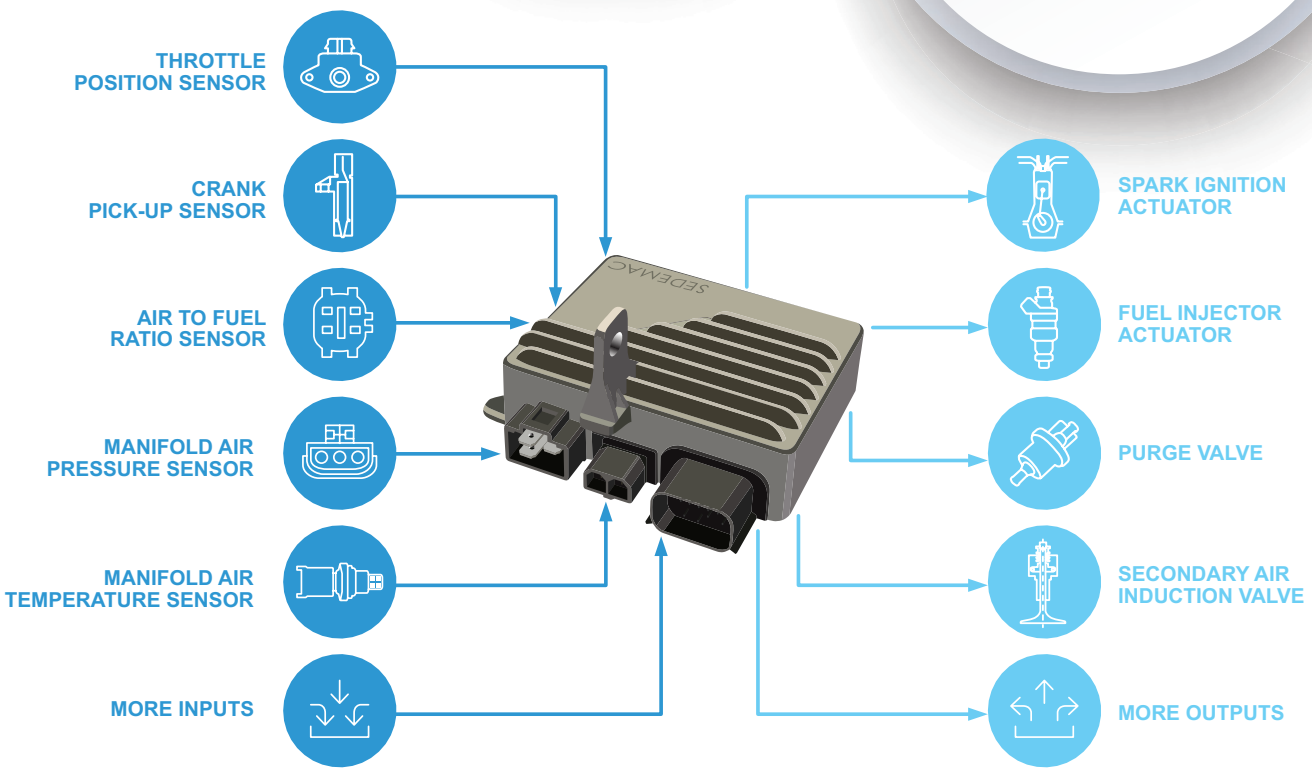
Integrated Starter Generator (ISG) is a technology that eliminates the need for conventional starter-motor based engine start system. In addition to a delightful start which is instant and smooth, **ISG** ensures improved reliability, and enables idle-stop-start feature. **ISG** system typically draws lesser current (vis-à-vis starter motor system) thereby enhancing improvement in battery life.

Conventional ISG technologies employ hall effect sensors which are exposed to hostile high temperature conditions (e.g. in motorcycles). Use of these sensors leads to reliability concerns, as well as need bigger electrical machines to accommodate them. **SEDEMAC's unique Sensorless ISG technology** eliminates the need for hall effect sensors and ensures **improved reliability** as well as **simpler, smaller electrical machines.**

ISG+ EFI ECU



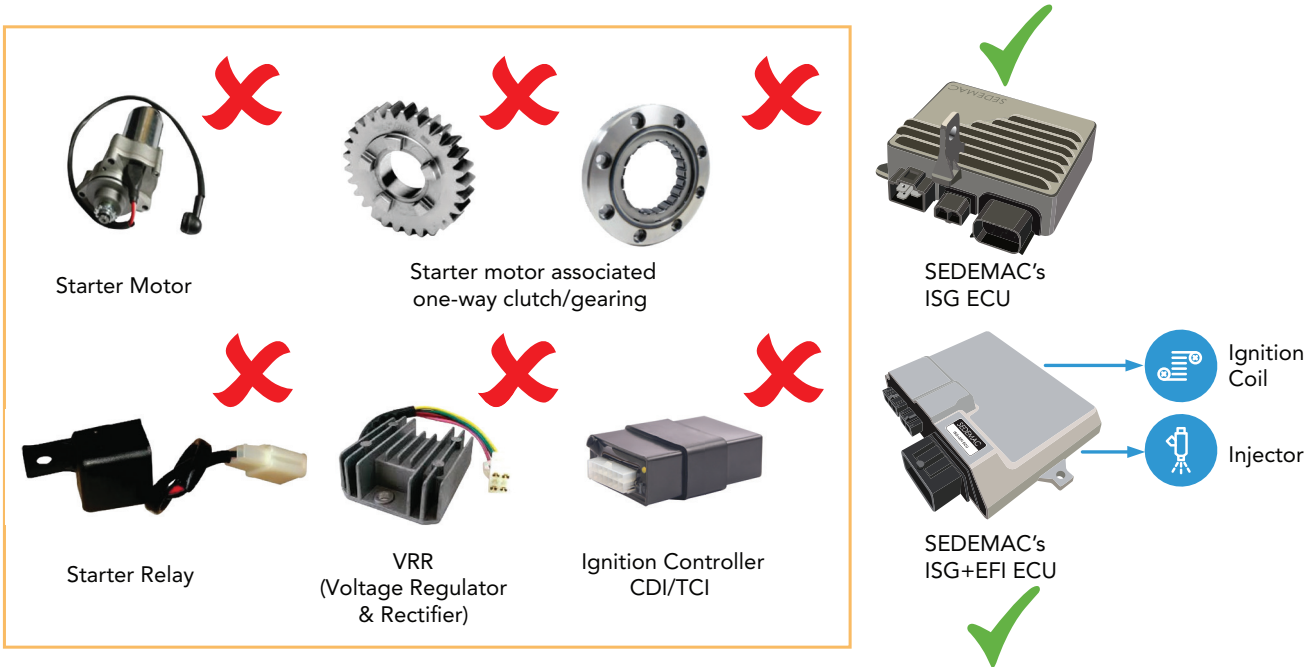
ISG ECU



Integration Possibilities: The forward-looking **ISG ECU** architecture enables integration of electronics of other control intensive technologies such as Electronic Fuel Injection (EFI) and Automated Manual Transmission (AMT).

ISG+EFI ECU

SEDEMAC's ISG+EFI technology integrates Sensorless ISG (Integrated Starter Generator), which eliminates the conventional starter-motor based engine start system, with **EFI** (Electronic Fuel Injection) system together in single **ECU.** Elimination of hall-effect sensor also results in simplified electric machine design.

This **ECU** can be used in 2 & 3 wheelers with crankshaft mounted alternator and provides instant and smooth engine start, along with cleaner exhaust emission, to meet stringent emission norms. Integration of two technologies leads to avoid duplication of parts, which results in cost saving. This technology is integrated with the algorithms that deliver improved reliability in varied atmospheric conditions.

ISG eliminates all components of the starter motor system, as well as the ignition controller and battery charging controller. Typically, the following parts are eliminated: starter motor, one-way clutch/gearing, starter relay, ignition CDI/TCI, and VRR (voltage regulator). These parts get eliminated and are replaced by a single part – **SEDEMAC's Sensorless ISG ECU and SEDEMAC's Sensorless ISG+EFI ECU**